

# Speculative Self-Distillation enables Efficient Knowledge Internalization

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## TL;DR

**Problem.** Self-distillation teaches a model a document by training it to match itself with the document in context. **on-policy** rollouts are accurate but spend most tokens where the teacher adds little.

**Method.** **SSD**: the student writes the rollout; the teacher writes only **inflection tokens**, where the two models disagree by more than  $\tau$ .

**Result.** **SSD** matches **on-policy** accuracy with **45% fewer supervised tokens** on three tasks and **forgets less**.

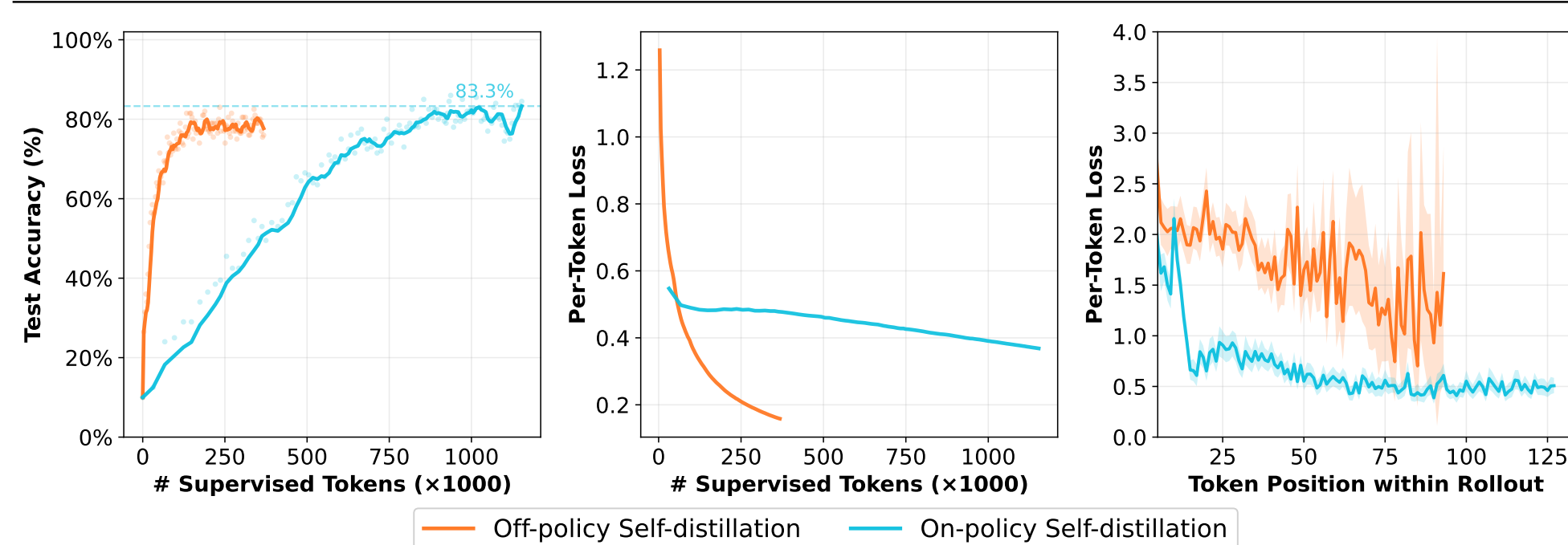
## Self-Distillation

- **Teacher**  $\pi_T(\cdot | x, C)$ : the model with privileged context  $C$  (a document). **Student**  $\pi_\theta(\cdot | x)$ : the same model without  $C$ .

$$\mathcal{L}(\theta) = \mathbb{E}_{y \sim \pi_{\text{gen}}} \left[ \frac{1}{|y|} \sum_t D(\pi_T(\cdot | x, C, y_{<t}) \| \pi_\theta(\cdot | x, y_{<t})) \right]$$

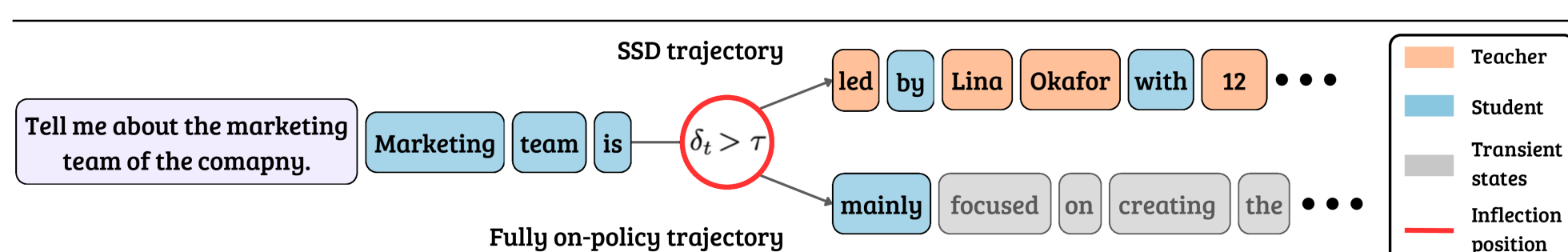
- Methods differ only in the rollout policy  $\pi_{\text{gen}}$ :
  - **off-policy** ( $\pi_{\text{gen}} = \pi_T$ ): strong signal, but trains on states the student never visits.
  - **on-policy** ( $\pi_{\text{gen}} = \pi_\theta$ , SDFT): no train-test mismatch, but slow.
- **SSD** uses a **per-token mix** of the two.

## Why On-Policy Is Slow



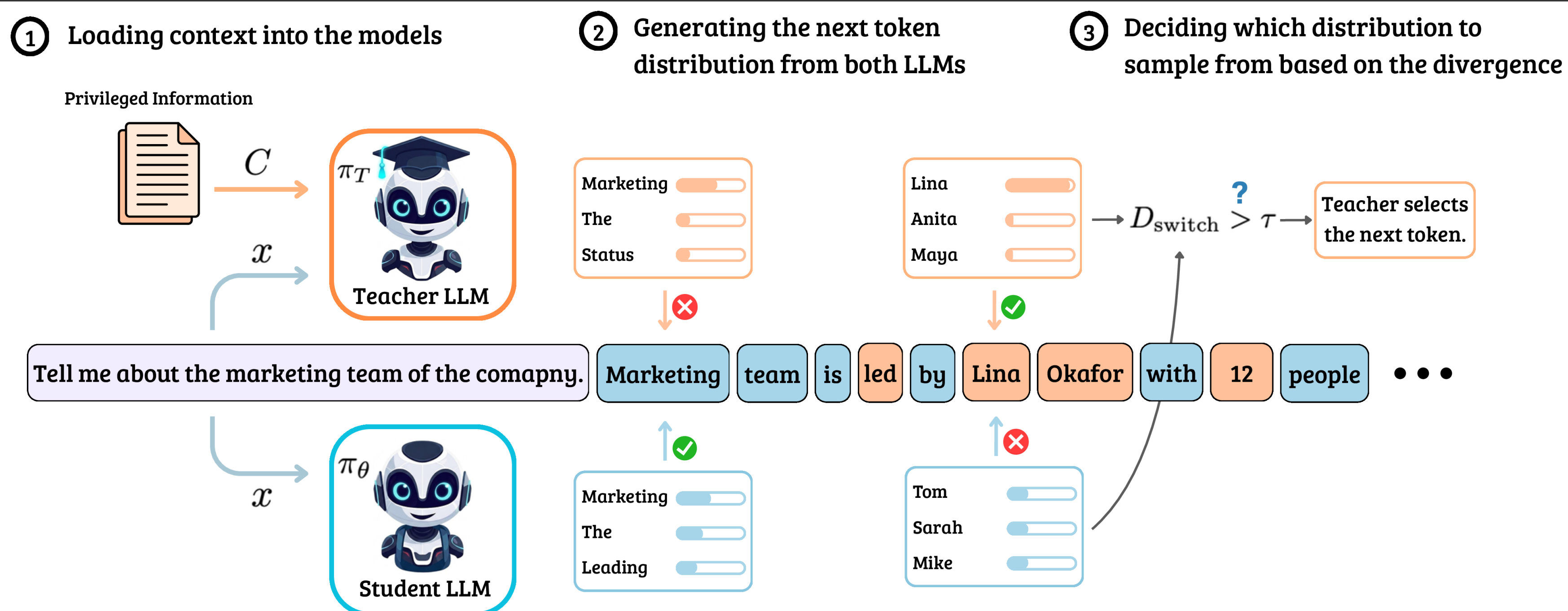
- **Signal** = per-token loss  $D(\pi_T \| \pi_\theta)$ : how much the student can still learn from that token.
- **off-policy** learns  $\sim 5\times$  faster but plateaus lower.
- Along **on-policy** rollouts the per-token signal **collapses after a few tokens**.

## Inflection Tokens



- Where the document changes the next token ( $\delta_t > \tau$ ), the teacher writes.
- **on-policy** keeps the student's token and drifts into **transient states**.

## Speculative Self-Distillation (SSD)



- Student generates by default.
- If  $D_{\text{switch}}(\pi_\theta \| \pi_T) > \tau$ , the teacher picks the token.
- Every position is trained with the same loss.
- $\tau \rightarrow 0$  is **off-policy**,  $\tau \rightarrow \infty$  is **on-policy**.

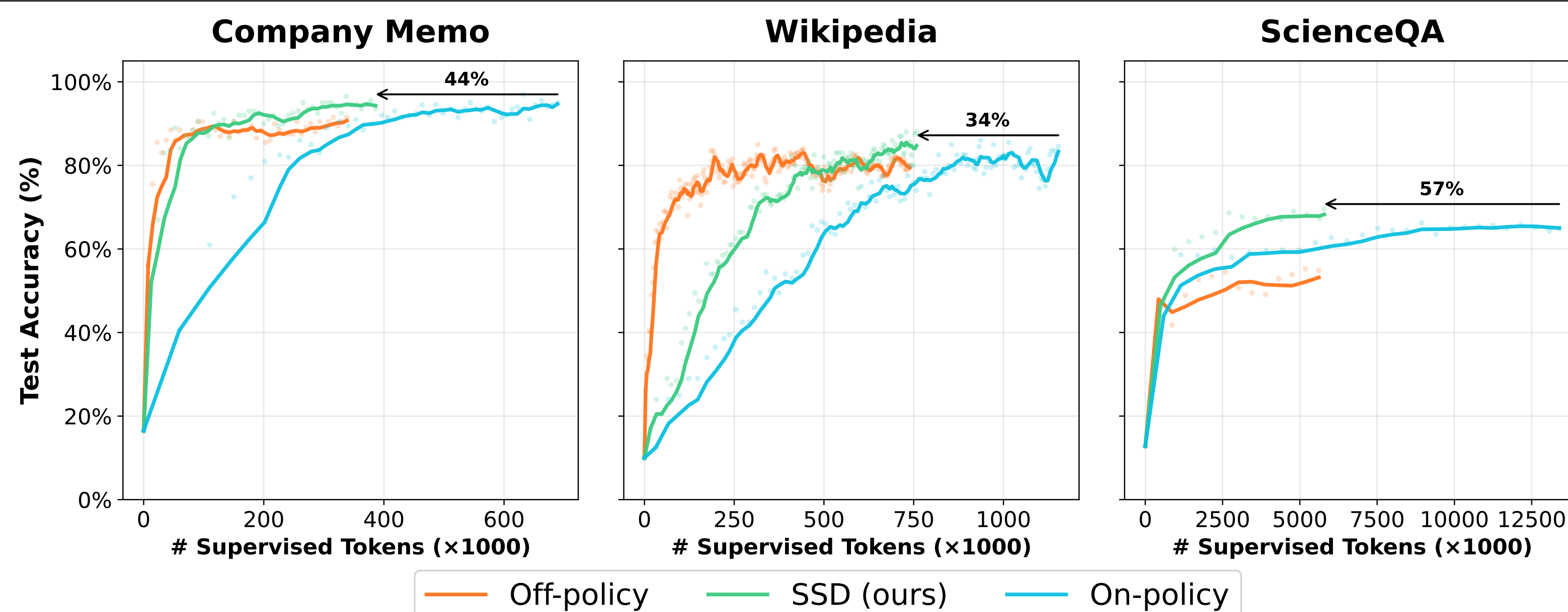
## Setup

**Model.** Qwen3-4B-Instruct as student and frozen teacher.

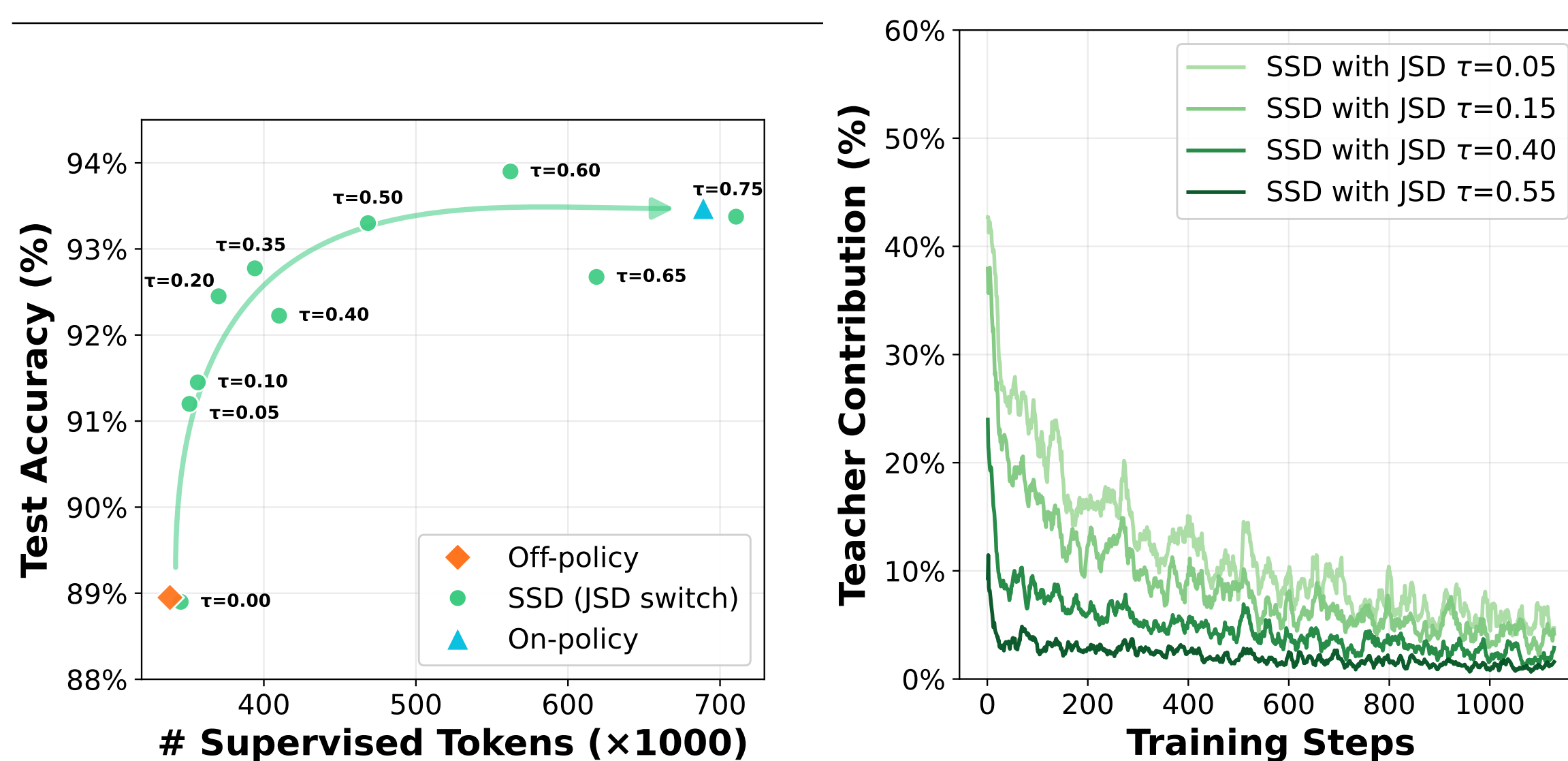
**Tasks.** Closed-book QA on a private memo, a post-cutoff Wikipedia article, and SciKnowEval chemistry.

**Cost.** Supervised tokens. Baselines: **off-policy** SFT/FKL, **on-policy** SDFT.

## SSD Reaches On-Policy Accuracy with 34–57% Fewer Tokens



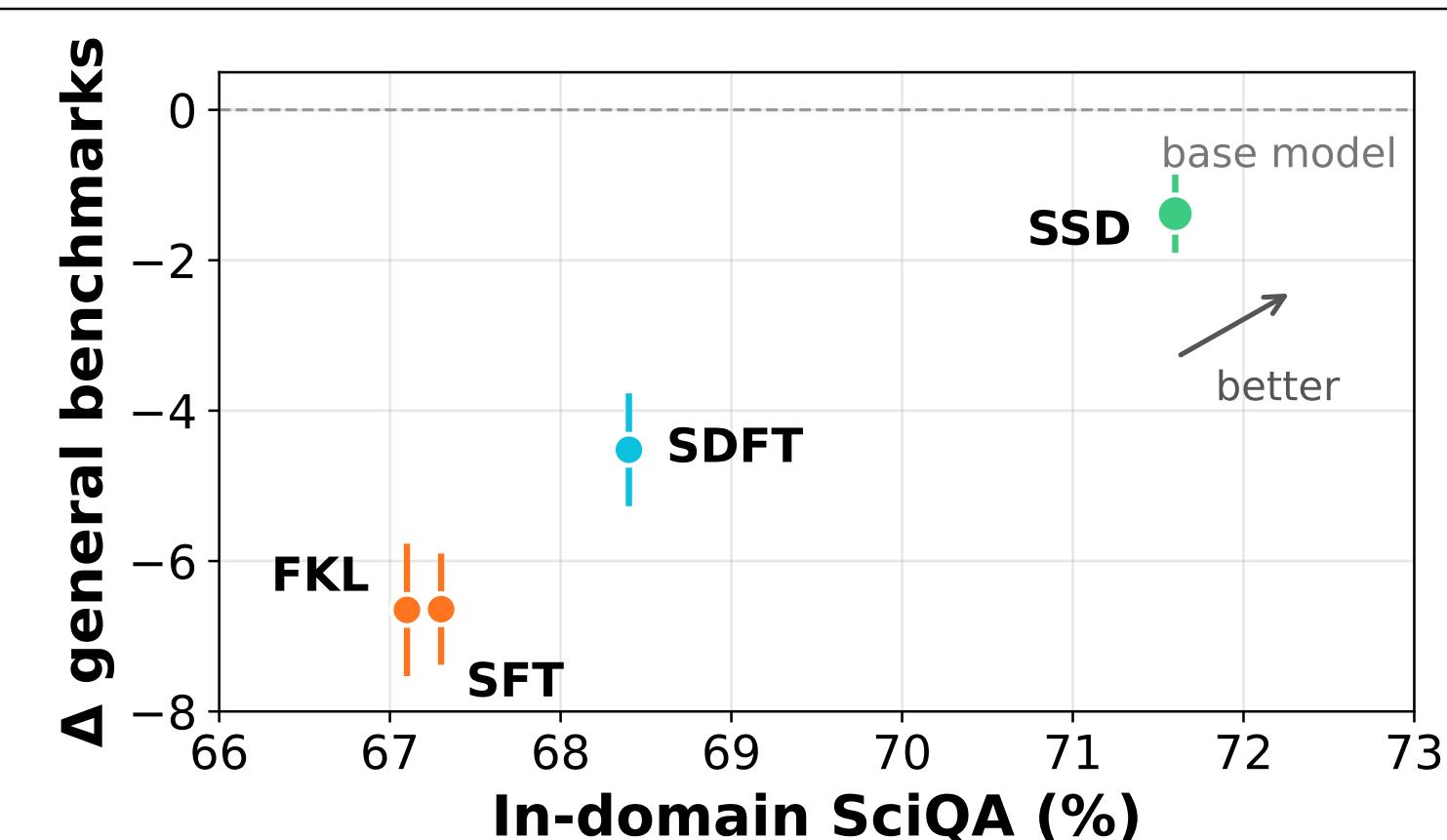
## One Knob, and a Built-in Curriculum



$\tau \in [0.2, 0.5]$  sits on the Pareto frontier.

The teacher's share fades with no schedule.

## Less Forgetting



Qwen2.5-7B, SciKnowEval chemistry: best in-domain accuracy, smallest drop on six general benchmarks.